The BAR Information Bottleneck

Self-calibration (sandwich variance), $O(1)$ differentiation, a Fisher–resistance correspondence, an estimation–detection conservation law, and a chirality completeness theorem

working notes: proofs for Paper 1 (v3, corrected)

Setup. Two thermodynamic states A, B (units $k_BT{=}1$). Forward works $\{W_i^f\}_{i=1}^{n_f} \sim P_f$, reverse works $\{W_j^r\}_{j=1}^{n_r} \sim P_r$. Unified coordinates $x_i = W_i^f$ (forward), $x_j = -W_j^r$ (reverse). By Crooks $P_f(x)/P_r(x) = e^{x-\Delta F}$, so the classes differ by a logistic tilt of intercept ΔF. With $M = \ln(n_f/n_r)$, $\sigma(z) = (1+e^{-z})^{-1}$, $p(x;d) = \sigma(x-d+M)$, the Bennett estimator $\widehat{\Delta F}$ is the MLE of the intercept (slope fixed at 1) under the *stratified two-sample* design (fixed n_f, n_r).

Lemma 1 (Score, curvature, and overlap)**.**

$$S(d) = \sum_{j\in r} p(x_j) - \sum_{i\in f}(1-p(x_i)), \qquad \partial_d S = -\sum_{all} p(1-p) =: -I(d).$$

$I(d) > 0$ (each $p(1-p) \in (0, \frac{1}{4}]$), so ℓ is strictly concave and the root $\widehat{\Delta F}$ is unique. I is the curvature *of the objective (the "overlap": large iff many $p \approx \frac{1}{2}$). It plays the role of the bread in the sandwich variance and the scale of the backward pass; it is* not *in general the inverse variance (see Theorem 2).*

Proof. Stationarity is the Bennett equation; $\partial_d^2\ell = -\sum p(1-p)$ via $\sigma' = \sigma(1-\sigma)$ and $\partial_d(x-d+M) = -1$. □

Theorem 1 ($O(1)$ differentiable backward; information-share gradient)**.** *With $\widehat{\Delta F}(\theta)$ defined by $S(\widehat{\Delta F}, \theta) = 0$ and $\partial_d S = -I \neq 0$, the implicit function theorem gives $d\widehat{\Delta F}/d\theta = \frac{1}{I}\,\partial_\theta S$. For $\theta = \mu^f$ ($x_i = \mu^f + \xi_i$) and $\theta = \mu^r$ ($x_j = -(\mu^r + \eta_j)$),*

$$\frac{d\widehat{\Delta F}}{d\mu^f} = \frac{I^f}{I}, \qquad \frac{d\widehat{\Delta F}}{d\mu^r} = -\frac{I^r}{I}, \qquad \left|\tfrac{d\widehat{\Delta F}}{d\mu^f}\right| + \left|\tfrac{d\widehat{\Delta F}}{d\mu^r}\right| = 1.$$

The backward pass is one division by the curvature I. (Unaffected by the variance correction below: it depends only on the deterministic second derivative.)

Proof. $\partial_{\mu^f}\big[-(1-p(x_i))\big] = p(1-p)$ gives $\partial_{\mu^f} S = I^f$; $\partial_{\mu^r} x_j = -1$ gives $\partial_{\mu^r} S = -I^r$. Shares sum to one since $I^f + I^r = I$. □

Remark 1 (envelope variant). If the loss is the BAR log-likelihood itself, $\partial_d \ell = 0$ at the root kills the implicit term (envelope theorem); the IFT gradient is needed only when $\widehat{\Delta F}$ feeds a downstream loss.

Remark 2 (numerical check). On $x^f = \{0, 1\}$, $x^r = \{-0.5\}$, $M = 0$: $\widehat{\Delta F} = -0.59571$, $I^f/I = 0.596825$ matches the IFT gradient to machine precision; $\partial_d S = -I$ to machine precision (Wolfram Language).

Assumption 1. (A1) Correct specification: predicted P_f, P_r equal the truth. (A2) Standard M-estimation regularity.

Theorem 2 (Self-calibration via the *sandwich* variance)**.** *Under Assumption 1, $\widehat{\Delta F}$ is an M-estimator with asymptotic variance*

$$\mathrm{Var}(\widehat{\Delta F}) = A^{-1}BA^{-1} = \frac{B}{I^2}, \qquad A = -\mathbb{E}[S'] = I, \qquad B = \mathrm{Var}\big(S(\Delta F)\big) = n_f\,\mathrm{Var}_f[p] + n_r\,\mathrm{Var}_r[p].$$

The information equality $A = B$ (which would give $\mathrm{Var} = 1/I$*) holds for prospective logistic regression but* fails *under the stratified two-sample design of BAR; the naive $1/I$ over-estimates the variance. The sandwich B/I^2 is the correct, computable-from-samples uncertainty (it coincides with the Bennett 1976 / Shirts–Chodera estimator), so calibration remains free, via the sandwich. It reduces to $1/I$ only in the information-equality (good-overlap) limit.*

Proof. $\widehat{\Delta F}$ solves $S = 0$; the one-step expansion $\widehat{\Delta F} - \Delta F = S(\Delta F)/I(\tilde{d})$ and the delta method give $\mathrm{Var}(\widehat{\Delta F}) = \mathrm{Var}(S)/I^2$ to leading order, with $\mathrm{Var}(S) = B$ by independence of the two samples and $\mathrm{Var}[1-p] = \mathrm{Var}[p]$. □

Remark 3 (numerical check). Correctly specified Gaussian model, $n_f = n_r = 20$, 2000 replicates: empirical $\mathrm{SD}(\widehat{\Delta F}) = 0.160$; mean sandwich se $= 0.159$ (ratio 0.994); mean naive $1/\sqrt{I} = 0.354$ (ratio 2.21). The sandwich is essentially exact at this N; the naive formula is off by 2.2× in se (Wolfram Language).

Proposition 1 (Finite-sample reliability and coverage)**.** *Let $\widehat{V} = \widehat{B}/\widehat{I}^2$ be the plug-in sandwich. With probability $\geq 1-\delta$, $|\widehat{\Delta F} - \Delta F| = O\big(\sqrt{\log(1/\delta)/N}\big)$ and $|\widehat{V}/V - 1| = O\big(\sqrt{\log(1/\delta)/N}\big)$, using Hoeffding on the bounded score/info/variance summands and the self-Lipschitz bound $|I'(d)| \leq I(d)$ (since $|1-2p| \leq 1$). The studentized statistic obeys*

$$\Big|\,\mathbb{P}\big((\widehat{\Delta F} - \Delta F)/\sqrt{\widehat{V}} \leq z\big) - \Phi(z)\,\Big| \;\leq\; \frac{0.56}{\sqrt{B}} = O(1/\sqrt{N}),$$

by Berry–Esseen, using $\mathbb{E}|X-\mu|^3 \leq$ Var *for $X \in [0,1]$ so the standardized third moment is $\leq 1/\sqrt{\mathrm{Var}\,S} = 1/\sqrt{B}$.*

Remark 4 (honest scope). This is the sampling variance under correct physics. Force-field misspecification adds a bias the frozen layer cannot self-correct (handled by a learned Δ-residual head with its own epistemic σ); network error about (μ, σ) is epistemic (ensemble). Total Belief variance $= B/I^2 + \sigma^2_{\mathrm{ens}} + \sigma^2_{\Delta}$.

Theorem 3 (Fisher–resistance correspondence; gauge-invariant acquisition)**.** *Relative measurements $y_e = (\varphi_j - \varphi_i) + \varepsilon_e$, $\varepsilon_e \sim \mathcal{N}(0, V_e)$ with $V_e = B_e/I_e^2$ the per-edge sandwich variance. The Gaussian posterior precision of φ is the weighted Laplacian*

$$L = \sum_{e=(i,j)} w_e\,(e_i - e_j)(e_i - e_j)^\top, \qquad w_e = \frac{1}{V_e} = \frac{I_e^2}{B_e},$$

i.e. the edge conductances are the inverse sandwich variances *(equal to the overlaps I_e only at the information-equality limit). Then:*

(i) $\mathrm{Var}(\varphi_b - \varphi_a) = (e_b - e_a)^\top L^+ (e_b - e_a) = \Omega_{ab}$ *(effective resistance).*

(ii) A single edge has $\Omega = V_e$, recovering the BAR sandwich variance.

(iii) Adding a measurement along c with precision g updates the contrast variance in $O(1)$, $\Omega' = \Omega/(1 + g\Omega)$ (Sherman–Morrison); KG equals the decision-weighted Ω-reduction.

(iv) $\ker L = \mathrm{span}\{$*component indicators*$\} \supseteq \mathrm{span}\{\mathbf{1}\}$*; contrasts in* $\ker L$ *have infinite variance and zero reduction, so* $\mathrm{KG} = 0$*. The acquisition is automatically gauge-invariant.*

Proof. The negative log-posterior is $\frac{1}{2}\sum_e w_e(y_e - (\varphi_j - \varphi_i))^2$ up to the gauge; its Hessian is L, posterior covariance L^+ on the range, giving (i)–(ii). (iii) is Sherman–Morrison on $c^\top(\cdot)^+c$. For (iv), all relative measurements are invariant under per-component shifts, so those directions lie in $\ker L$. □

Remark 5 (numerical check). Triangle with conductances $(1, 0.5, 2)$: $\Omega_{12} = 0.714286$ equals the series–parallel value; $\ker L = \mathrm{span}\{\mathbf{1}\}$; adding edge $(1,2)$ with $g = 0.7$ gives exact $\Omega' = 0.476190$, matching $\Omega/(1 + g\Omega)$ to machine precision (Wolfram Language).

Corollary 1 (the unified objects, corrected)**.** *The overlap* I_e *governs* $\{$*curvature, backward-pass scale (Thm. 1)*$\}$*. The precision* I_e^2/B_e *governs* $\{$*calibration (Thm. 2), Laplacian weight (Thm. 3)*$\}$*. These coincide (a single object,* $V = 1/I$*) only at the information-equality limit. They diverge* most in the high-overlap regime *(*I/B *grows large as* $p \to \frac{1}{2}$*; numerically* $I/B \approx 17$ *at* 0.5σ *separation), so the naive* $1/I$ *over-estimates the variance most for the well-overlapped, low-variance edges, converging to the sandwich only as overlap* $\to 0$*. The sandwich is therefore essential precisely on the* reliable*, well-overlapped edges an active learner exploits. Naive* $1/I$ *would systematically distrust the best edges (up to* $\sim$13$\times$ *inflation on real FEP edges).* (Direction corrected; verified three ways: controlled MC, population I/B, real FEP edges; see `docs/results_figA.md`.)

Estimation–detection conservation law

Setup. Collect the per-edge relative measurements of Theorem 3 into $y = D\varphi + \varepsilon$, where $D \in \mathbb{R}^{E\times N}$ is the signed incidence matrix (row $e = (i,j)$ equals $\mathbf{e}_j^\top - \mathbf{e}_i^\top$), $\varepsilon \sim \mathcal{N}(0, W^{-1})$, $W = \mathrm{diag}(w_e)$, $w_e = 1/V_e$. Whitening by $W^{1/2}$ gives design $\tilde{D} = W^{1/2}D$; the GLS fit has orthogonal projector (hat matrix) $H = \tilde{D}L^+\tilde{D}^\top = W^{1/2}DL^+D^\top W^{1/2}$, with $L = D^\top WD$ the weighted Laplacian (Theorem 3), and residual-maker $M = I_E - H$ (reusing M for the residual-maker, as in the main text; the log-count ratio of the estimator setup does not appear here). The closure statistic is $X^2 = z^\top Mz$ in whitened coordinates $z = W^{1/2}y$.

Theorem 4 (Estimation–detection conservation law)**.** *Let the* curl-leverage *be* $h_e = M_{ee}$*. Then*

(i) $h_e + w_e\Omega_e = 1$*, with* Ω_e *the effective resistance across edge e (Theorem 3);*

(ii) $\sum_e h_e = \mathrm{trace}\, M = E - \mathrm{rank}\, D = E - N + c = \mathrm{dof}$*, the number of independent cycles (first Betti number; c connected components);*

(iii) under a systematic shift of size μ *on edge e (the mean of y shifted by* $\mu\,\mathbf{1}_e$*),* X^2 *is noncentral* χ^2_{dof} *with noncentrality* $\lambda_e = h_e\,\mu^2/V_e$*.*

Hence a bridge edge (e in no cycle) has $h_e = 0$ *and is undetectable at every* μ*; a node-consistent bias* $b = D\psi$ *maps into* $\mathrm{range}(\tilde{D}) = \mathrm{range}(H)$ *and is annihilated by* M*, so the test is blind to it* by construction *rather than by assumption; and the smallest shift reaching unit noncentrality is* $\delta_e^* = \sqrt{V_e/h_e}$*.*

Proof. (i) $H_{ee} = w_e\,(\mathbf{e}_j - \mathbf{e}_i)^\top L^+(\mathbf{e}_j - \mathbf{e}_i) = w_e\Omega_e$ by Theorem 3(i); hence $h_e = M_{ee} = 1 - H_{ee} = 1 - w_e\Omega_e$. (ii) H is the orthogonal projector onto $\mathrm{range}(\tilde{D})$, so $\mathrm{trace}\, H = \mathrm{rank}\,\tilde{D} = \mathrm{rank}\, D = N - c$

($W^{1/2}$ is a full-rank positive diagonal, and rank $D = N-c$ for a graph with c components); therefore trace $M = E-(N-c)$, the cycle-space dimension. (iii) $z \sim \mathcal{N}(\nu, I_E)$ since $\mathrm{Cov}(y) = W^{-1}$; under the null $\nu \in \mathrm{range}(\tilde{D})$, so $M\nu = 0$ and $X^2 \sim \chi^2_{\mathrm{dof}}$. Shifting the mean of y by $\mu\,\mathbf{1}_e$ moves ν by $s = \mu W^{1/2}\mathbf{1}_e = (\mu/\sqrt{V_e})\,\mathbf{1}_e$, giving noncentrality $\|Ms\|^2 = s^\top M s = (\mu^2/V_e)\,M_{ee} = h_e\mu^2/V_e$. Finally $h_e = \|M\mathbf{1}_e\|^2$ (idempotent M), so $h_e = 0 \Leftrightarrow \mathbf{1}_e \in \mathrm{range}(H)$, i.e. $\mathbf{1}_e$ lies in the cut space range(D) (e in no cycle); $W^{1/2}D\psi \in \mathrm{range}(\tilde{D})$ gives the gradient-bias annihilation; and δ_e^* solves $\lambda_e = 1$. □

Remark 6 (numerical check). Across the 48 OpenFE IndustryBenchmarks2024 systems with ≥ 1 independent cycle, $\sum_e h_e =$ dof holds to a maximum deviation 1.78×10^{-14}, and the two evaluations of h_e (from $M = I-H$ and from $1-w_e\Omega_e$) agree to machine precision; `make figLev`.

Chirality completeness

Setup. A configuration is a point cloud $\{(p_i, z_i)\}$ in $\mathbb{R}^3$. $SO(3)$ are proper rotations ($\det R = +1$); $O(3)$ adds reflections ($\det R = -1$). The enantiomer of M is $M' = \sigma M$ for a reflection σ. M is chiral iff M' is not a proper-rigid-motion image of M.

Lemma 2 ($O(3)$-invariant readouts collapse enantiomers). *If f is $O(3)$-invariant then $f(M') = f(\sigma M) = f(M)$. Any function of pairwise distances $\|p_i - p_j\|$ (which are $O(3)$-invariant) is constant on the pair $\{M, M'\}$; hence a distance-only network is provably enantiomer-blind.*

Theorem 5 (A parity-odd pseudoscalar separates enantiomers). *Let $\chi(M) = \det[\,p_2-p_1, p_3-p_1, p_4-p_1\,]$. For any linear A, $\chi(A\cdot M) = \det(A)\,\chi(M)$. Hence χ is $SO(3)$-invariant ($\det R = +1$) and parity-odd: $\chi(M') = -\chi(M)$. For chiral M (non-coplanar points) $\chi(M) \neq 0$, so $\chi(M') = -\chi(M) \neq \chi(M)$ and χ distinguishes the enantiomers.*

Proof. $\det[Av_1, Av_2, Av_3] = \det(A)\det[v_1, v_2, v_3]$ with $v_k = p_{k+1} - p_1$; $\det(\sigma) = -1$. □

Corollary 2 (completeness: one lost bit). *For generic (chiral, full-rank) configurations, the pair $\big(\{\|p_i - p_j\|\},\ \mathrm{sgn}\,\chi\big)$ is a complete $SO(3)$-invariant. Pairwise distances determine the configuration up to $O(3)$, i.e. up to $\{M, \sigma M\}$; $\mathrm{sgn}\,\chi$ takes opposite signs on the two and selects the enantiomer. Thus $O(3)$-invariance loses exactly the chirality bit, and a pseudoscalar restores it. Here "complete" means complete for the enantiomer-pair (1-bit) problem on a configuration already pinned by its full set of pairwise distances; the sign of a single triple product does not by itself characterize the chirality of an arbitrary >4-atom molecule, which in general requires a consistent set of pseudoscalars.*

Remark 7 (implementation). The pseudoscalar is the parity-odd ℓ=0 irrep (`0o`), realized as the triple product $v_1\cdot(v_2\times v_3) = \det[v_1, v_2, v_3]$, an odd contraction of vector features. A readout must contain a `0o` channel; a parity-even readout lies in the kernel and is blind. Note that $\|v\|$ and $v_i v_j$ are $O(3)$-invariant (even), so vector features pooled into norms/dot-products remain enantiomer-blind: the *odd* triple product is required.